



An auxiliary diagnostic tool for common fundus diseases based on fundus color photography and light-weight classification models

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Abstract

Objective To evaluate the performance of two lightweight neural network models in the diagnosis of common fundus diseases and make comparison to another two classical models.

Methods A total of 16,000 color fundus photography were collected, including 2000 each of glaucoma, diabetic retinopathy (DR), high myopia, central retinal vein occlusion (CRVO), age-related macular degeneration (AMD), optic neuropathy, and central serous chorioretinopathy (CSC), in addition to 2000 normal fundus. Fundus photography was obtained from patients or physical examiners who visited the Ophthalmology Department of Beijing Tongren Hospital, Capital Medical University. Each fundus photography has been diagnosed and labeled by two professional ophthalmologists. Two classical classification models (ResNet152 and DenseNet121), and two lightweight classification models (MobileNetV3 and ShufflenetV2), were trained. Area under the curve (AUC), sensitivity, specificity, accuracy, positive predictive value, and negative predictive value were used to evaluate the performance of the four models.

Results Compared with the classical classification model, the total size and number of parameters of the two lightweight classification models were significantly reduced, and the classification speed was sharply improved. Compared with the DenseNet121 model, the ShufflenetV2 model took 50.7% less time to make a diagnosis on a fundus photography. The classical models performed better than lightweight classification models, and Densenet121 showed highest AUC in five out of the seven common fundus diseases. However, the performance of lightweight classification models is satisfying. The AUCs using MobileNetV3 model to diagnose AMD, diabetic retinopathy, glaucoma, CRVO, high myopia, optic atrophy, and CSC were 0.805, 0.892, 0.866, 0.812, 0.887, 0.868, and 0.803, respectively. For ShufflenetV2model, the AUCs for the above seven diseases were 0.856, 0.893, 0.855, 0.884, 0.891, 0.867, and 0.844, respectively.

Conclusion The training of light-weight neural network models based on color fundus photography for the diagnosis of common fundus diseases is not only fast but also has a significant reduction in storage size and parameter number compared with the classical classification model, and can achieve satisfactory accuracy.

Keywords Eye fundus diseases · Lightweight neural network model · Auxiliary diagnosis

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Key messages

What is known:

- The total size and number of parameters of classical models are of huge volume.

What is new:

- The application scenarios for the auxiliary diagnosis of fundus diseases are mostly community hospitals, usually lacking professional ophthalmologist and the hardware configuration is low. Therefore, reducing the computational burden is the only choice.
- Lightweight classification models has good practical value and high efficiency.

Introduction

In the past decade, convolutional neural network (CNN) has been widely used in image classification [1, 2], image segmentation [3–5], and object detection [6, 7], such as diagnosis of fundus diseases. The accuracy of CNN for image classification has been widely verified. As early as 2016, some researchers used CNN to train diagnostic models for diabetic retinopathy (DR) and diabetic macular edema, and the sensitivity and specificity reached 97.5% [1] and 93.4%, respectively. In 2019, a comprehensive diagnostic model for fundus diseases was developed using 494,661 fundus photographs of patients with glaucoma, DR or age-related macular degeneration (AMD), which achieved a diagnostic sensitivity of 100% [2] for severe DR.

One problem with CNN is that from the early days of AlexNet [8], to VGG [9], GoogLeNet [10], ResNet [11], DenseNet [12], and so on. The number of network layers keeps increasing, from 7 to 16, to 22, to 152, and even more. Some models use millions of parameters to improve the accuracy [13, 14], at a cost of efficiency. Efficiency involves two aspects: the storage of models and the speed for disease diagnosis. Sadly, the above “large” models require large device memory, as well as a high device configuration to ensure a fast respond speed, typically in milliseconds.

For the auxiliary diagnosis of fundus diseases, the application scenarios are mostly community hospitals, where there are usually no professional ophthalmologist and the hardware configuration is low. Therefore, reducing the computational burden is the only choice. The light-weight neural network model is designed as a more efficient “network computing mode” (mainly for the convolutional layer), so as to reduce network parameters with little or no loss in network performance. The aim of this study was to evaluate the accuracy of two

light-weight classification models for diagnosis of fundus diseases, and make comparison to another two classical models.

Methods and materials

Eye fundus images collection

A total of 16,000 fundus photography inspected between April, 2022 and June, 2023 in Beijing Tongren Hospital, Capital Medical University were included in the present study. Of these, 2000 were normal fundus (Fig. 1A) and 2000 were for each of the seven common fundus diseases: glaucoma (Fig. 1B), DR (Fig. 1C), high myopia (Fig. 1D), central retinal vein occlusion (CRVO) (Fig. 1E), AMD (Fig. 1F), optic neuropathy (Fig. 1G), and central serous chorioretinopathy (CSC) (Fig. 1H).

Diagnosis of eye fundus diseases

Glaucoma was defined using the definition developed by the International Society for Geographical and Epidemiological Ophthalmology (ISGEO) [15]. The presence of DR was assessed from the retinal fundus images using the modified Airlie House classification system for the Early Treatment of Diabetic Retinopathy [16]. Early AMD was defined as presence of either (1) large indistinct soft or reticular drusen, or (2) both large distinct soft drusen and retinal pigmentary abnormalities. Late AMD was defined as the presence of geographic atrophy (GA) or neurovascular AMD [17]. CRVO was comprehensively judged by retinal blood stasis, retinal hemorrhage, edema, and exudation. High myopia was comprehensively judged by the tessellated

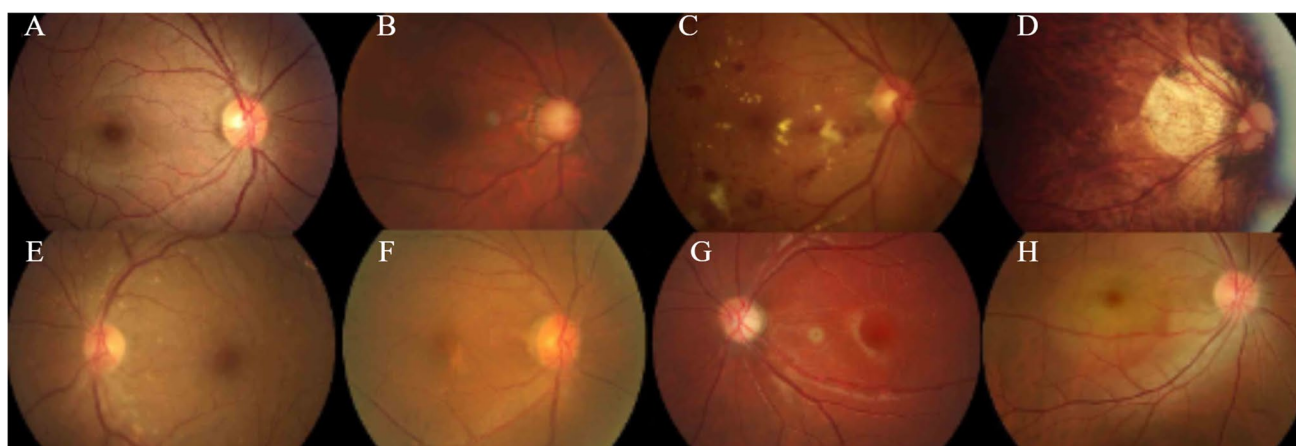


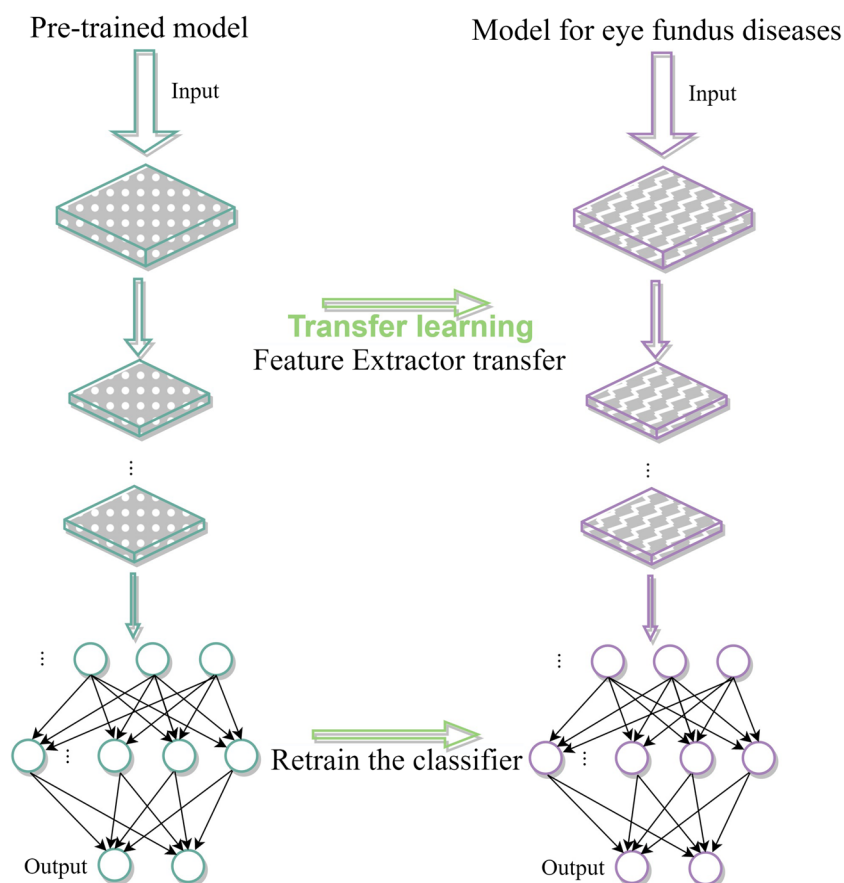
Fig. 1 Eye fundus used in the present study. (A Normal fundus; B glaucoma; C DR; D high myopia; E CRVO; F AMD; G optic neuropathy; H CSC)

fundus appearance, choroidal atrophy, and different degrees of atrophic changes in the macular region. The diagnosis of optic atrophy is based on visual field defect and optic disc color. The diagnosis of CSC is a comprehensive diagnosis based on whether there is macular edema, macular color, and whether the foveal reflex disappears.

Light-weight classification models

The light-weight classification models used in this study are MobileNetV3 (MobileNetV3_small) and ShuffleNetV2 (ShuffleNet_v2_x0_5). Both of them reduce the number of parameters and storage space requirement by designing more efficient convolution units. The

Fig. 2 Transfer learning for eye fundus diseases classification models in the present study



MobileNetV3 model combines the following four features: (1) The common convolution layer in the classic CNN model is replaced with the depth-wise separable convolution, which consists of the depth-wise convolution and point-wise convolution. (2) The inverted residual with linear bottleneck. (3) The squeeze-and-excite structure. (4) H-swishj activation function is used in the structure instead of Swish function. Based on the above improvements, the number of parameters and the computational cost of the model are reduced. ShuffleNet V2 [18] takes the actual reasoning speed as the guide, proposes the channel split operation, divides the input feature into two parts, and realizes the feature reuse effect similar to DenseNet model, so it balances the accuracy and speed of the model well.

In this study, we compared the model size, number of parameters, diagnostic speed, and accuracy of the two light-weight classification models [11, 12] (ResNet152, DenseNet121) with the other two classical classification models using the same dataset.

Transfer learning

In the present study, transfer learning was used to train and test the model (Fig. 2). During the training process, the parameters obtained after training on the ImageNet dataset (containing more than 14 million images and 1000 categories) [8] are loaded into the feature extraction part of the models. Only the classifier parameters are trained, and the number of categories is modified to meet the classification requirements of this study.

Model training, validation, and test

All patients' information was desensitized before training. The extra black edges were removed from the fundus images, and the size was adjusted to 128*128 (without changing the original proportion). A total of 16,000 images were randomly divided into training data-set,

validation data-set, and test set according to the ratio of 8:1:1. The number of epoch was set to be five.

In the model training process, the batch size was set to be 32 and there were 188 batches. The stochastic gradient descent optimizer was used, and cross entropy was chosen as the loss function. The initial learning rate is 0.1. After the training, the saved model was applied to the test data-set and the area under the curve (AUC) was calculated to represent the accuracy of the model.

Hardware configuration

The hardware configuration used in the model training and verification process in this study is as follows: The CPU is Intel (R) Core I7 6700 K, and the main frequency is 4.0 GHz. The graphics card is NVIDIA GeForce RTX 2060 with 12 GB of video memory. The memory size is 48 GB. The operating system is Windows 10. The deep learning framework used is PyTorch and the programming language is Python (version 3.10.4).

Table 2 Performance of ResNet152

Diseases	AUC	Sensitivity	Specificity	PPV	NPV
AMD	0.882	0.832	0.921	0.913	0.857
Diabetic retinopathy	0.911	0.866	0.945	0.936	0.879
Glaucoma	0.928	0.881	0.979	0.982	0.894
CRVO	0.941	0.907	0.987	0.983	0.906
High myopia	0.926	0.879	0.997	0.997	0.892
Optic atrophy	0.883	0.852	0.902	0.879	0.882
CSC	0.879	0.824	0.914	0.898	0.855

AMD, age-related macular degeneration. *CRVO*, central retinal vein occlusion. *CSC*, central serous chorioretinopathy. *AUC*, area under the curve. *PPV*, positive predictive value. *NPV*, negative predictive value

Table 3 Performance of DenseNet121

Diseases	AUC	Sensitivity	Specificity	PPV	NPV
AMD	0.883	0.839	0.924	0.911	0.866
Diabetic retinopathy	0.940	0.892	0.981	0.977	0.912
Glaucoma	0.878	0.821	0.936	0.919	0.855
CRVO	0.925	0.879	0.975	0.968	0.894
High myopia	0.931	0.879	0.991	0.986	0.889
Optic atrophy	0.883	0.862	0.905	0.889	0.886
CSC	0.887	0.836	0.928	0.915	0.859

AMD, age-related macular degeneration. *CRVO*, central retinal vein occlusion. *CSC*, central serous chorioretinopathy. *AUC*, area under the curve. *PPV*, positive predictive value. *NPV*, negative predictive value

Table 1 Model size and number of parameters

CNN models	Total size of model (MB)	Parameters (millions)	Average time for making diagnosis (ms)
Classic models			
ResNet152	428	60	163
DenseNet121	27	8	205
Lightweight models			
MobileNetV3	21	2.5	72
ShufflenetV2	14	1.4	101

Table 4 Performance of MobileNetV3

Diseases	AUC	Sensitivity	Specificity	PPV	NPV
AMD	0.805	0.733	0.879	0.846	0.781
Diabetic retinopathy	0.892	0.856	0.931	0.922	0.871
Glaucoma	0.866	0.811	0.905	0.883	0.829
CRVO	0.812	0.721	0.897	0.864	0.788
High myopia	0.887	0.853	0.916	0.906	0.875
Optic atrophy	0.868	0.859	0.873	0.845	0.866
CSC	0.803	0.757	0.855	0.821	0.804

AMD, age-related macular degeneration. CRVO, central retinal vein occlusion. CSC, central serous chorioretinopathy. AUC, area under the curve. PPV, positive predictive value. NPV, negative predictive value

Table 5 Performance of ShufflenetV2

Diseases	AUC	Sensitivity	Specificity	PPV	NPV
AMD	0.856	0.811	0.904	0.879	0.842
Diabetic retinopathy	0.893	0.853	0.922	0.912	0.877
Glaucoma	0.855	0.792	0.917	0.891	0.823
CRVO	0.884	0.831	0.909	0.894	0.853
High myopia	0.891	0.837	0.928	0.915	0.866
Optic atrophy	0.867	0.841	0.884	0.862	0.853
CSC	0.844	0.813	0.875	0.862	0.831

AMD, age-related macular degeneration. CRVO, central retinal vein occlusion. CSC, central serous chorioretinopathy. AUC, area under the curve. PPV, positive predictive value. NPV, negative predictive value

Statistical analysis

The open source R program (<https://www.r-project.org/>, Version 4.2.0) was used for statistical analysis. Receiver operating characteristic curve was used to analyze the diagnostic performance of the model. AUC, sensitivity, specificity, positive prediction value (PPV), and negative prediction value (NPV) were used to evaluate the performance of four models. The manual diagnosis results of ophthalmologists were taken as the gold standard, and $AUC > 0.75$ indicated high diagnostic value. The higher the AUC value, the sensitivity, specificity, PPV, and NPV, the higher the model performance.

Results

Compared with the classical classification models, the model size and parameter number of the light-weight models were significantly reduced. The ShufflenetV2 model had the smallest size and the least parameter number, while the ResNet152 model had the largest size and the most parameter number (Table 1). Both

the model size and the number of parameters of the two models differ by dozens of times. Besides, compared with the classical DenseNet121 model, the ShufflenetV2 model reduced the average classification time by more than 50%.

The AUCs of ResNet152 model in the diagnosis of AMD, DR, glaucoma, CRVO, high myopia, optic atrophy, and CSC were 0.882, 0.911, 0.928, 0.941, 0.926, 0.883, and 0.879, respectively (Table 2). The ResNet152 model had the highest sensitivity (0.907) in the diagnosis of CRVO and the highest specificity (0.997) in the diagnosis of high myopia.

The AUCs of DenseNet121 model in the diagnosis of AMD, DR, glaucoma, CRVO, high myopia, optic atrophy, and CSC were 0.883, 0.940, 0.878, 0.925, 0.931, 0.883, and 0.887, respectively (Table 3). The DenseNet121 model had the highest sensitivity (0.892) in the diagnosis of DR and the highest specificity (0.991) in the diagnosis of high myopia.

The AUCs of MobileNetV3 model in the diagnosis of AMD, DR, glaucoma, CRVO, high myopia, optic atrophy, and CSC were 0.805, 0.892, 0.866, 0.812, 0.887, 0.868, and 0.803, respectively (Table 4). The MobileNetV3 model had the highest sensitivity (0.859) in the diagnosis of optic atrophy and the highest specificity (0.931) in the diagnosis of DR.

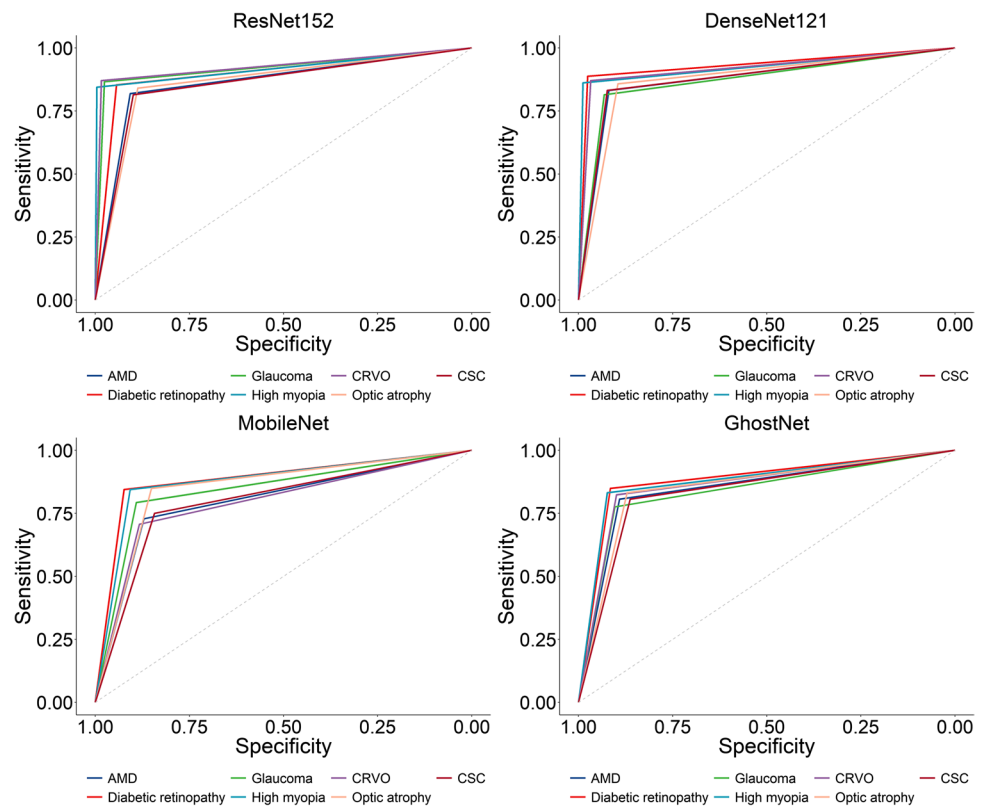
The AUCs of ShufflenetV2 model in the diagnosis of AMD, DR, glaucoma, CRVO, high myopia, optic atrophy, and CSC were 0.856, 0.893, 0.855, 0.884, 0.891, 0.867, and 0.844, respectively (Table 5). The ShufflenetV2 model had the highest sensitivity (0.853) in the diagnosis of DR and the highest specificity in the diagnosis of high myopia (0.928).

The weighted Kappa values of the ResNet152 model, the DenseNet121 model, the MobileNetV3 model, and the ShufflenetV2 model were 0.873, 0.896, 0.761, and 0.823, respectively. Overall, the accuracy of the light-weight classification models was slightly worse than that of the classical classification models, but still acceptable (Fig. 3).

Discussion

With the increasing maturity of CNN in the image field, the auxiliary diagnosis of fundus disease has gradually become possible. Some countries and regions have put the auxiliary diagnosis of fundus diseases into practice [19, 20]. In some developing countries, including China, primary hospitals are relatively short of ophthalmologists, and the hardware of primary hospitals, including computers, is often of low configuration, which can only meet the daily office needs. It is

Fig. 3 The ROC curves of classical classification models and light-weight classification models for fundus diseases diagnosing



difficult to run the model based on classical CNN training to carry out the diagnosis of fundus diseases. The establishment of a light-weight diagnostic tool with high accuracy is of great significance for improving the diagnostic ability of fundus diseases in primary hospitals and realizing the implementation of hierarchical diagnosis and treatment.

In the present study, the classical classification models perform better than the light-weight model, which is understandable because the classical classification models have more complex parameters. Accordingly, the model size and computation time are increased. In contrast, the light-weight models sacrifice a small amount of accuracy, but the model size and computation time are greatly improved. In the present study, compared with the classical DenseNet121 model and ResNet152 model, the light-weight models reduced the number of parameters and model size sharply, besides, the average time for making diagnosis was greatly reduced. Taking the classical DenseNet121 model and the light-weight MobileNetV3 model for example, the model size reduced by 1.3 times, the number

of parameters reduced by 3.2 times, and the average time for making diagnosis reduced from 205 to 72 ms. Meanwhile, The maximum loss of diagnostic accuracy was 12.3% depending on the fundus diseases. For optic atrophy and glaucoma, the loss of diagnostic accuracy was merely 2.7% and 3.1%. And most importantly, the sacrifice on accuracy was acceptable, since the AUCs of two light-weight models in diagnosis of seven fundus diseases were all above 0.8.

In general, our study shows that the light-weight auxiliary models for fundus diseases diagnosis trained by transfer learning showed satisfying accuracy. The strength of the present study is that we have trained a relatively comprehensive multi-classification tool covering common fundus diseases, and the sample size was relatively large. The limitation lies in that the sensitivity of MobileNetV3 light-weight model had a relatively lower sensitivity on the diagnosis for AMD and CRVO, compared to the diagnosis for other fundus diseases. It is worth paying attention to improve the feature learning ability of the two light-weight models for the two diseases in the future.

Author contribution JD W and MR L did the data analysis and wrote the manuscript. ML L and RZ advised and supported the study design and conduction. JD W and CX C conducted outcomes measurements. KC investigated the study, revised the manuscript, and provided the study conceptual.

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Data availability The data that support the findings of this study are available on request from the corresponding author.

Declarations

Ethics approval This study was approved by the Ethics Committee, Beijing Tongren Hospital, Capital Medical University.

Consent for publication Not applicable.

Competing interests The authors declare no competing interests.

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